

Student	Hours_Studied	Attendance (%)	Assignments_Submitted	Previous_Score	Marks
Amit	2	75	4	32	35
Ravi	4	80	5	45	50
Sita	6	85	6	58	65
Priya	8	90	7	72	80
Rahul	10	95	8	88	95
Neha	5	82	5	52	60
Arjun	7	88	6	68	75
Pooja	3	78	4	40	45
Karan	9	92	7	80	88
Meera	6	86	6	62	70

1. Write Python code to display:

- Number of rows and columns
- Column names
- Data types of all columns

2. Calculate:

- Mean, median, and standard deviation of **Marks**
- Mean attendance
- Maximum and minimum Previous_Score

3. Find:

- Student with the highest Marks
- Student with the second-highest Marks
- Difference between their scores

4. Determine which feature among:

- Hours_Studied
- Attendance
- Assignments_Submitted
- Previous_Score

has the strongest correlation with Marks and Provide the correlation values.

5. Display all students who satisfy both conditions:

- Attendance $\geq 85\%$
- Marks ≥ 70

Also calculate their average Marks.

6. Create a new feature:

Performance_Index = (Marks + Previous_Score)/2

Find:

- Student with the highest Performance_Index
- Average Performance_Index of the class

7. Create:

- Bar chart: Student vs Marks